

Cheng Xu

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Research Interests: VLA Models, Robotic Manipulation, World Models, Multimodal Perception, and Tactile Sensing

Education

The University of Tokyo

M.Eng. in Electrical Engineering and Information Systems

Current research: Vision-based landmark recognition and multi-sensor fusion for train localization in autonomous train operation.

Tokyo, Japan

Apr. 2026 – Present

University of Sussex

B.Eng. (Hons) in Robotics and Electrical Engineering, First-Class Honours

Dual-degree program average: **88/100** | Rank: **Top 5%**

Brighton, United Kingdom

Sep. 2021 – Jun. 2025

Zhejiang Gongshang University

B.Eng. in Electronic Information Engineering

Dual-degree program average: **88/100** | Rank: **Top 5%**

Hangzhou, China

Sep. 2021 – Jun. 2025

Selected Experience

AGIBOT Japan

Engineering Assistant Intern

- Assisted engineers with the disassembly and maintenance inspection of G2 humanoid robots during on-site technical support.
- Reviewed the operating requirements of a planned robotic towel-loading task and conducted an on-site assessment of the working environment.

Tokyo, Japan

Jun. 25 – Jul. 3, 2026

Selected Projects

Tactile Sensors for Robotic Manipulators

- Fabricated and experimentally evaluated an eight-layer, 16×16 flexible piezoresistive tactile sensor array using Velo-stat as the sensing material and an Arduino Nano for row-column signal acquisition.
- Implemented a row-column readout circuit and PCB layout in EasyEDA, and integrated an Arduino-Python pipeline for serial data acquisition and real-time pressure-map visualization.
- Conducted load-response and contact-distribution experiments, compared multiple noise-reduction methods, and selected a four-stage pipeline.

Oct. 2024 – Mar. 2025

VLA Fundamentals

- Conducted a line-by-line analysis of Physical Intelligence's official π_0 PyTorch implementation, tracing tensor shapes and data flow through multimodal preprocessing, prefix and suffix embeddings, attention-mask construction, flow-matching training, and iterative action denoising.
- Ongoing work: preparing to reproduce and fine-tune $\pi_{0.5}$ based on research papers for robotic manipulation tasks.

May 2026 – Present

Technical Skills

Programming & ML: Python, PyTorch, C, MATLAB

Hardware & Tools: Arduino, EasyEDA, Multisim, schematic capture, PCB layout

Honors & Awards

- Outstanding Student Award (Top 1%),** Zhejiang Gongshang University, 2024 – 2025
- First-Class Scholarship,** University of Sussex, 2024 – 2025
- Academic Scholarship,** University of Sussex, 2022 – 2023

Languages

Japanese: JLPT N1

English: TOEFL iBT 87